



# Four Fixes on Base, No Migration

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Fifty lines of Solidity make Morpheus settlement real, with no chain move. It is the cheapest entry in this series.

**The problem, in their numbers.** MOR trades at \$1.95, market cap near \$17M [verified]. Real revenue is Lido and Aave yield, about \$476K a year and falling, down from \$3.24M a quarter at the 2024 launch [verified]. Emission outruns real revenue near ten to one [verified]. This is not a fraud: principal is withdrawable one to one. The honest problem is dilution outrunning revenue, over a settlement number no one can check. Settlement is `pricePerSecond` times elapsed time [verified]. The provider node signs a seven-field receipt, including the input and output token counts that say how much inference happened (`abi.go:12-19`) [verified]. The contract decodes five of the seven and drops the two that matter (`SessionRouter.sol:237`) [verified]. Only the provider signs (`SessionRouter.sol:227`). Nothing stops a provider opening a session against its own bid (`SessionRouter.sol:106`). The owner key, an anonymous five-of-nine, can rewrite any facet in one transaction, with no timelock [verified]. Morpheus built the honest half. The node already signs the work. The contract discards it.

**The fix.** Four edits to the `SessionRouter` facet in `Morpheus-Lumerin-Node/smart-contracts/`. None touch the token, the treasury, the staking math, or the chain.

- **Decode the whole receipt.** Add two `uint32` fields to the decode tuple at `SessionRouter.sol:237`. Settlement then reads the tokens, not just the clock. About five lines.
- **Make the consumer co-sign.** `closeSession` already holds `session.user` and the recover-and-compare helper (`SessionRouter.sol:543`). Run it against the consumer, and a provider cannot post volume the consumer will not sign.
- **Forbid self-dealing.** One `require(session.user != bid.provider)` removes the wash trade where a provider funds a session against its own bid.
- **Timelock the owner.** A timelock between the Safe and the owner role turns a stolen key into an announced upgrade with a withdrawal window, not an instant silent loss.

**The outcome.** About fifty lines of Solidity [illustrative: sum of the four edits above], on the diamond that already exists. No migration, no new token, no vote to move capital. Best case: settlement reads work the node already signs, both parties sign the bill, self-dealing is blocked, and a compromised owner key buys a window instead of the treasury.

**Assumptions.** Reading `inputTokens` proves a count was signed, not that the tokens were produced; attesting the work is a later step (LP-313). The receipt still settles on the provider's reported count until a consumer disputes it. A timelock narrows the owner key's blast radius; it does not remove it. This makes the base honest, not complete.

**Relevance.** Lux built `mord`, an un-fakeable settlement engine where faked work earns zero and model substitution is slashed; forty tests pass as of 2026-08-29 [verified]. We know which fifty lines, and why. This fix is the first rung, and the rest of the series stands on an honest receipt.

ADOPT-09. Facet `SessionRouter` in `Morpheus-Lumerin-Node/smart-contracts/`. Diamond `0x6aBE1d282f72B474E54527D93b979A4f64d3030a` on Base. Citations checked against the working tree. Leads to LP-313 and LP-323. • `research@lux.network`